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Molding apparatus

Abstract

An apparatus for molding process includes a mount including a base, a support on the base, first and second side columns, and a mounting member moveably on tops of the first and second columns; pneumatic first and second half molding assemblies for forming a mold; and first and second auxiliary devices cooperating with a die to insert the mold for molding. There are further provided second cooling devices, first and second lifting devices, first and second lowering devices, and a control device.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The invention relates to molding and more particularly to a molding apparatus having improved characteristics.

2. Description of Related Art

(2) Conventional molding apparatuses are disadvantageous because they are complicated in components and prone to malfunction; quality of products is poor; and may hurt employees in operation if sufficient care is not taken.

(3) Thus, the need for improvement still exists.

SUMMARY OF THE INVENTION

(4) It is therefore one object of the invention to provide a molding apparatus comprising a mount including a base, a support provided on a center of a top of the base, first and second columns provided on two sides of the top of the base, and a mounting member moveably provided on tops of the first and second columns; a first half molding assembly provided on the base and between the support and the first column, the first half molding assembly including a first half-circular member having a first half sprue, and a first pneumatic cylinder contacting the first half-circular member for moving the first half-circular member; a second half molding assembly provided on the base and between the support and the second column, the second half molding assembly including a second half-circular member having a second half sprue, and a second pneumatic cylinder contacting the second half-circular member for moving the second half-circular member wherein the first and second half-circular members are configured to move toward each other form a circular mold on the support, the first and second half sprues form a sprue of the circular mold, and the first and second pneumatic members are configured to activate in synchronism to separate the circular mold; a die provided on a center of the mounting member and including a block at an end and a pneumatic cylinder downward extending to upward or downward move the block; a first auxiliary device provided on an underside of the mount and including a first lateral moving member for moving the first auxiliary device toward the die or away from the die, a first lifting member adjacent to the die, a first L-shaped arm provided on the first lifting member and configured to move upward to engage with the die or downward to disengage from the die, and a first shaping member provided on an underside of the first L-shaped arm; a second auxiliary device provided on the underside of the mount and including a second lateral moving member for moving the second auxiliary device toward the die or away from the die, a second lifting member adjacent to the die, a second L-shaped arm provided on the second lifting member and configured to move upward or downward, and a second shaping member provided on an underside of the second L-shaped arm wherein the first and second shaping members and the block are configured to move into the circular mold; first and second cooling devices are provided in the base under the first half molding assembly and the second half molding assembly respectively for cooling the circular mold; first and second lifting devices provided on two sides of a top of the mounting member respectively wherein the first and second lifting devices are configured to move the mounting member along the first and second columns; first and second lowering devices provided on two sides of an underside of the base respectively wherein the first and second lowering devices are configured to lift or lower the base; and a control device electrically connected to the first and second half molding assemblies, the die, the first and second auxiliary devices, the first and second

cooling devices, the first and second lifting devices, and the first and second lowering devices respectively.
(5) The above and other objects, features and advantages of the invention will become apparent from the following detailed description taken with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a schematic front view of a molding apparatus according to a first preferred embodiment of the invention;
(2) FIG. **2** schematically depicts opening and closing of the half molding assemblies;
(3) FIG. **3** is a schematic perspective view of the die;
(4) FIG. **4** schematically depicts the auxiliary devices;
(5) FIG. **5** schematically depicts the forming of a complete tool consisting of the die and the shaping members;
(6) FIG. **6** schematically depicts the tool to be inserted into the circular mold;
(7) FIG. **7** is a perspective view of the tool inserted into the circular mold;
(8) FIG. **8** is a view similar to FIG. **1** showing movements of the auxiliary devices;
(9) FIG. **9** is a view similar to FIG. **8** showing movement of the die;
(10) FIG. **10** is a view similar to FIG. **9** showing a joining of the auxiliary devices and the die;
(11) FIG. **11** is a view similar to FIG. **10** showing movements of the pneumatic cylinders of the half molding assemblies;
(12) FIG. **12** is a view similar to FIG. **11** showing movements of the lowering devices;
(13) FIG. **13** is a view similar to FIG. **12** showing the tool consisting of the die and the shaping members inserted into the circular mold;
(14) FIG. **14** is a longitudinal sectional view showing the molten metal being poured into the sprue in a molding process;
(15) FIG. **15** is a view similar to FIG. **13** showing the tool consisting of the die and the shaping members disengaged from the circular mold;
(16) FIGS. **16**, **17**, **18** and **19** are views similar to FIG. **15** showing steps of returning the components to their original positions after the molding process respectively;
(17) FIG. **20** is a block diagram of the molding apparatus; and
(18) FIG. **21** schematically depicts a molding apparatus according to a second preferred embodiment of the invention.
DETAILED DESCRIPTION OF THE INVENTION
(19) Referring to FIGS. **1** to **20**, a molding apparatus in accordance with a first preferred embodiment of the invention comprises a mount **10**, two half molding assemblies **20**, **20**A, a die **30**, two auxiliary devices **40**, **40**A, two cooling devices **50**, two lifting devices **60**, two lowering devices **70**, and a control device **80** as detailed below.
(20) The mount **10** includes a base **11**, a support **12** provided on a center of a top of the base **11**, two columns **13** provided on two sides of the top of the base **11**, and a mounting member **14** moveably provided on tops of the columns **13**.
(21) The half molding assembly **20**A is provided on the base **11** and between the support **12** and the left column **13**. The half molding assembly **20** is provided on the base **11** and between the support **12** and the right column **13**. The half molding assembly **20**A includes a half-circular member **21**A and a pneumatic cylinder **22**A contacting the half-circular member **21**A for moving the half-circular member **21**A. The half molding assembly **20** includes a half-circular member **21** and a pneumatic cylinder **22** contacting the half-circular member **21** for moving the half-circular member **21**. The half-circular members **21**, **21**A may move toward each other form a circular mold **21**B on the

support **12**. The circular mold **21**B has a top opening **21**C. The half-circular member **21**A has a half sprue **23**A and the half-circular member **21** has a half sprue **23**. The half sprues **23**A, **23** form a sprue **23**B of the circular mold **21**B.

(22) The die **30** is provided on a central portion of the mounting member **14** and includes a block **31** at an end and a pneumatic cylinder **32** downward extending to upward or downward move the block **31**.

(23) The auxiliary device **40** is provided on an underside of the mount **10** and includes a lateral moving member **41** for moving the auxiliary device **40** toward the die **30** or away from the die **30**, a lifting member **42** adjacent to the die **30**, an L-shaped arm **43** provided on the lifting member **42** and configured to move upward or downward, and a shaping member **44** provided on an underside of the L-shaped arm **43** and having an arcuate groove **441**. The auxiliary device **40**A is provided on an underside of the mount **10** and includes a lateral moving member **41**A for moving the auxiliary device **40**A toward the die **30** or away from the die **30**, a lifting member **42**A adjacent to the die **30**, an L-shaped arm **43**A provided on the lifting member **42**A and configured to move upward to engage with the die **30** or downward to disengage from the die **30**, and a shaping member **44**A provided on an underside of the L-shaped arm **43**A and having an arcuate groove **441**A.

(24) In operation, the lifting members **42**, **42**A activate in synchronism to move the shaping member **44**, **44**A toward the block **31** and clamp the block **31** via the L-shaped arms **43**, **43**A. The shaping members **44**, **44**A and the block **31** together move downward to insert into the circular mold **21**B. A piston ring **45** can be placed in the grooves **441**, **441**A.

(25) The cooling device **50** is provided in the base **11** under the half molding assembly or the half molding assembly **20**A for cooling the circular mold **21**B in operation.

(26) The lifting devices **60** are provided on two sides of a top of the mounting member **14** respectively. In operation, the lifting devices **60** activate to move the mounting member **14** downward along the columns **13** until the shaping members **44**, **44**A and the block **31** together insert into the circular mold **21**B. Alternatively, the lifting devices **60** activate to move the mounting member **14** upward along the columns **13** to move the shaping members **44**, **44**A and the block **31** away from the circular mold **21**B.

(27) The lowering devices **70** are provided on two sides of an underside of the base **11** respectively. In an operation, the lowering devices **70** activate to lift the base **11** to finish the molding. After the molding, the lowering devices **70** activate to lower the base **11** to its inoperative position and the pneumatic members **22**, **22**A activate in synchronism to separate the circular mold **21**B.

(28) The control device **80** is electrically connected to the half molding assemblies **20**, **20**A, the die **30**, the auxiliary devices **40**, **40**A, the cooling devices **50**, the lifting devices **60**, and the lowering devices **70** respectively. Thus, the control device **80** may open or close the half molding assemblies **20**, **20**A, lower or lift the die **30**, move and lift or lower the auxiliary devices **40**, **40**A, and activate the cooling devices **50**, the lifting devices **60**, and the lowering devices **70** in a molding process.

(29) Specifically, in the molding process molten metal is pouring into the circular mold **21**B through the sprue **23**B. Also, the cooling devices **50** activate to cool the circular mold **21**B until a product **100** is finished. Thereafter, the pneumatic cylinder **32** activates to lift the die **31**, the lowering devices **70** activate to lower the base **11** to its inoperative position, the pneumatic members **22**, **22**A activate in synchronism to separate the circular mold **21**B, and the lifting devices **60** activate so that the lifting members **42**, **42**A may activate in synchronism to move the shaping member **44**, **44**A upward via the L-shaped arms **43**, **43**A. Finally, the product **100** can be removed.

(30) Referring to FIG. **21**, a molding apparatus in accordance with a second preferred embodiment of the invention is shown. The characteristics of the second preferred embodiment are substantially the same as that of the first preferred embodiment except the following: for manufacturing a piston **101**, a wearing ring **90** is provided on the support **12**. The wearing ring **90** can be clamped by the half-circular members **21**, **21**A so that a wearing part **102** can be formed on a lower portion of the piston **101**.

(31) A longer piston can be manufactured when a distance between the base **11** and the mounting member **14** is longer and a shorter piston can be manufactured when a distance between the base **11** and the mounting member **14** is shorter.
(32) While the invention has been described in terms of preferred embodiments, those skilled in the art will recognize that the invention can be practiced with modifications within the spirit and scope of the appended claims.

Claims

1. A molding apparatus, comprising: a mount including a base, a support disposed on a center of a top of the base, first and second columns disposed on two sides of the top of the base, and a mounting member moveably disposed on tops of the first and second columns; a first half molding assembly disposed on the base and between the support and the first column, the first half molding assembly including a first half-circular member having a first half sprue, and a first pneumatic cylinder contacting the first half-circular member for moving the first half-circular member; a second half molding assembly disposed on the base and between the support and the second column, the second half molding assembly including a second half-circular member having a second half sprue, and a second pneumatic cylinder contacting the second half-circular member for moving the second half-circular member wherein the first and second half-circular members are configured to move toward each other form a circular mold on the support, the first and second half sprues form a sprue of the circular mold, and the first and second pneumatic members are configured to activate in synchronism to separate the circular mold; a die disposed on a center of the mounting member and including a block at an end and a pneumatic cylinder downward extending to upward or downward move the block; a first auxiliary device disposed on an underside of the mount and including a first lateral moving member for moving the first auxiliary device toward the die or away from the die, a first lifting member adjacent to the die, a first L-shaped arm disposed on the first lifting member and configured to move upward to engage with the die or downward to disengage from the die, and a first shaping member disposed on an underside of the first L-shaped arm; a second auxiliary device disposed on the underside of the mount and including a second lateral moving member for moving the second auxiliary device toward the die or away from the die, a second lifting member adjacent to the die, a second L-shaped arm disposed on the second lifting member and configured to move upward or downward, and a second shaping member disposed on an underside of the second L-shaped arm wherein the first and second shaping members and the block are configured to move into the circular mold; first and second cooling devices are disposed in the base under the first half molding assembly and the second half molding assembly respectively for cooling the circular mold; first and second lifting devices disposed on two sides of a top of the mounting member respectively wherein the first and second lifting devices are configured to move the mounting member along the first and second columns; first and second lowering devices disposed on two sides of an underside of the base respectively wherein the first and second lowering devices are configured to lift or lower the base; and a control device electrically connected to the first and second half molding assemblies, the die, the first and second auxiliary devices, the first and second cooling devices, the first and second lifting devices, and the first and second lowering devices respectively.
2. The molding apparatus of claim 1, wherein the circular mold is open to a top.
3. The molding apparatus of claim 1, wherein the first and second shaping members each include an arcuate groove.